

Assignment 4

1. Consider a RDBMS team working to store the records of a newly established school in your area. They have identified the id (7), name (50), mobile (11), guardian's email (24) and address (40) as the fields of records. Considering an overhead of 20% on data volume with initial 1200 records and a growth rate of 900 records per 6 months, **calculate** the expected volume of the database table in 4 years.
2. A company needs a program to produce payroll of the employees. Here the main module (Produce payroll) will receive a control data Valid payroll transaction from Get Valid Timesheet module. Then Get Master module receives employee ID from Produce payroll and returns employee data to it. Upon receiving employee data, Produce payroll module forward it to Calculate employee pay module and receive employee pay information. After that Produce payroll and send this information to Generating Accounting entities, Produce employee pay cheque and Informing bank for transfer module. Here Generate Accounting Entities is off page connector and Produce Employee pay cheque is ON page connection. Produce payroll can call anyone from Produce employee pay cheque and Informing bank for transfer at a time. **Design a Structure chart from the above scenario.**
3. **Compare** File based storage and Relational DBMS.
4. Suppose you are developing a school management system. The client has requested to add a feature where a student can check his/her current GPA and also the cumulative gpa. Now you put the codes necessary for both of this to work in the same module. Now point out **which cohesion** is taking place here and if there is scope to improve then suggest how can it be done?